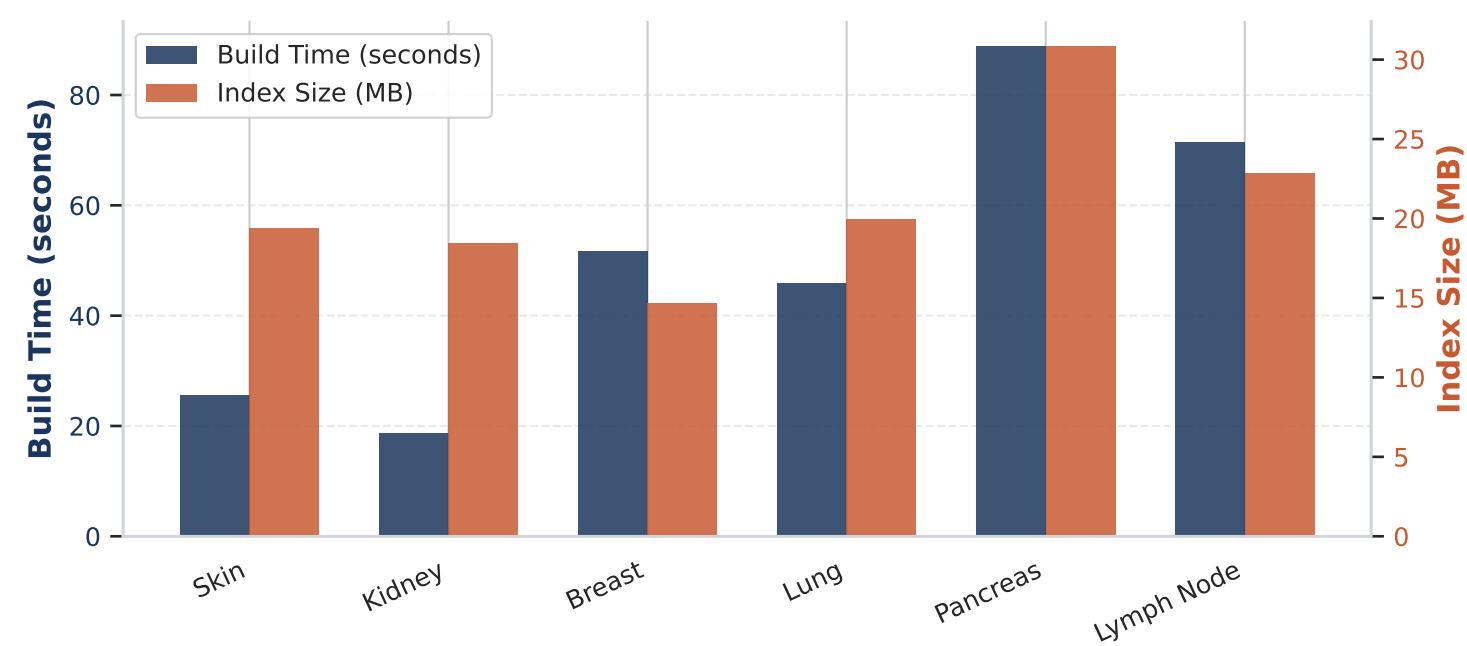


Figure 1: SPINDLE Architecture and Performance

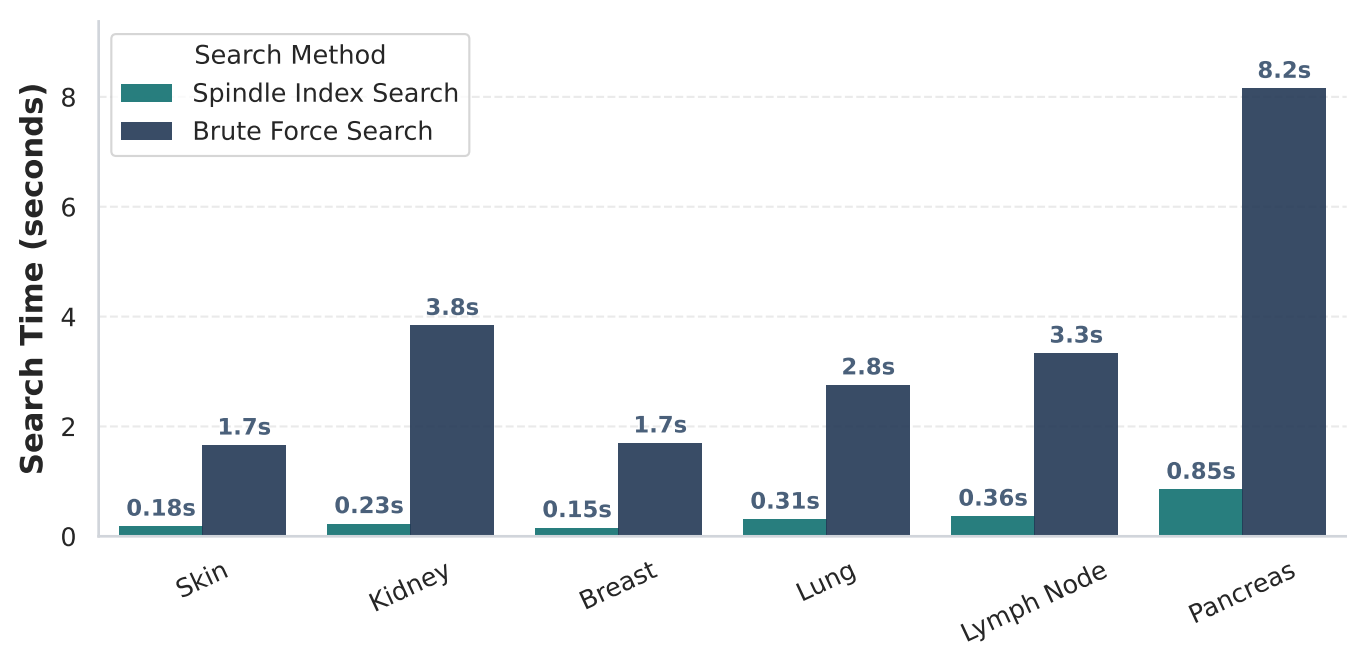
A

Index Scalability and Memory Footprint



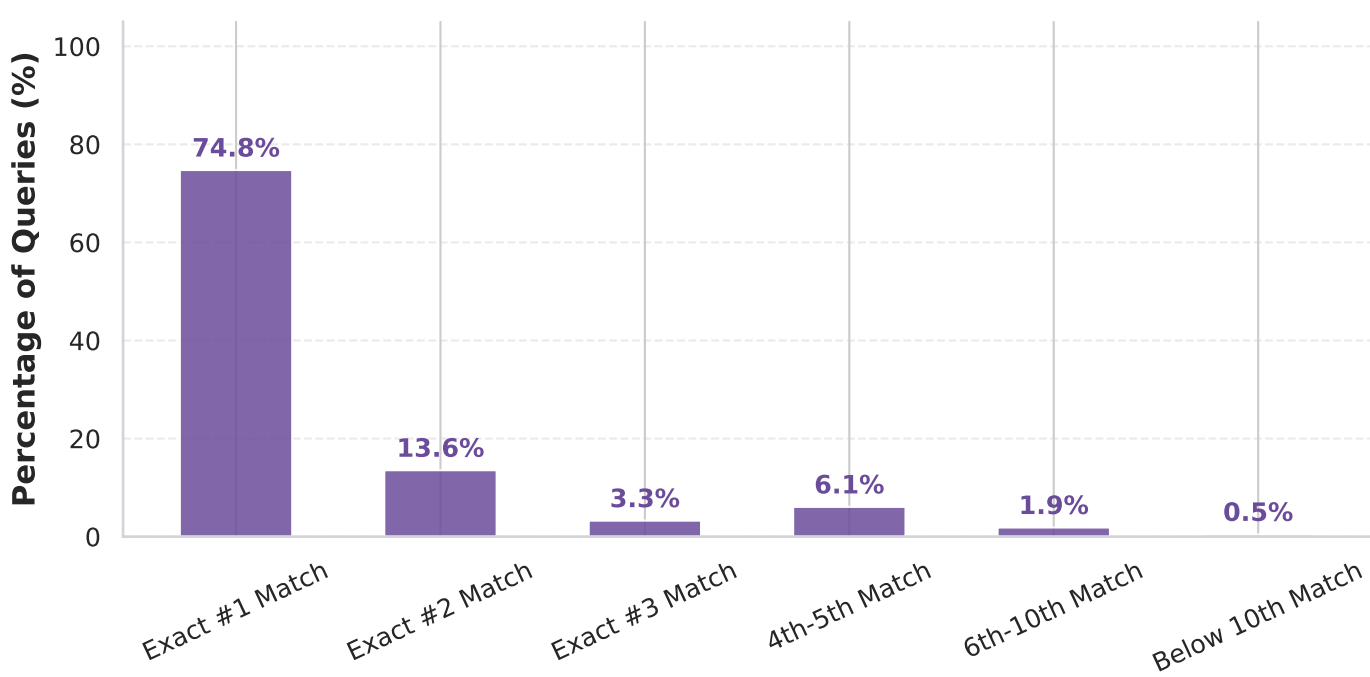
B

Query Execution Time (Spindle vs Brute Force)



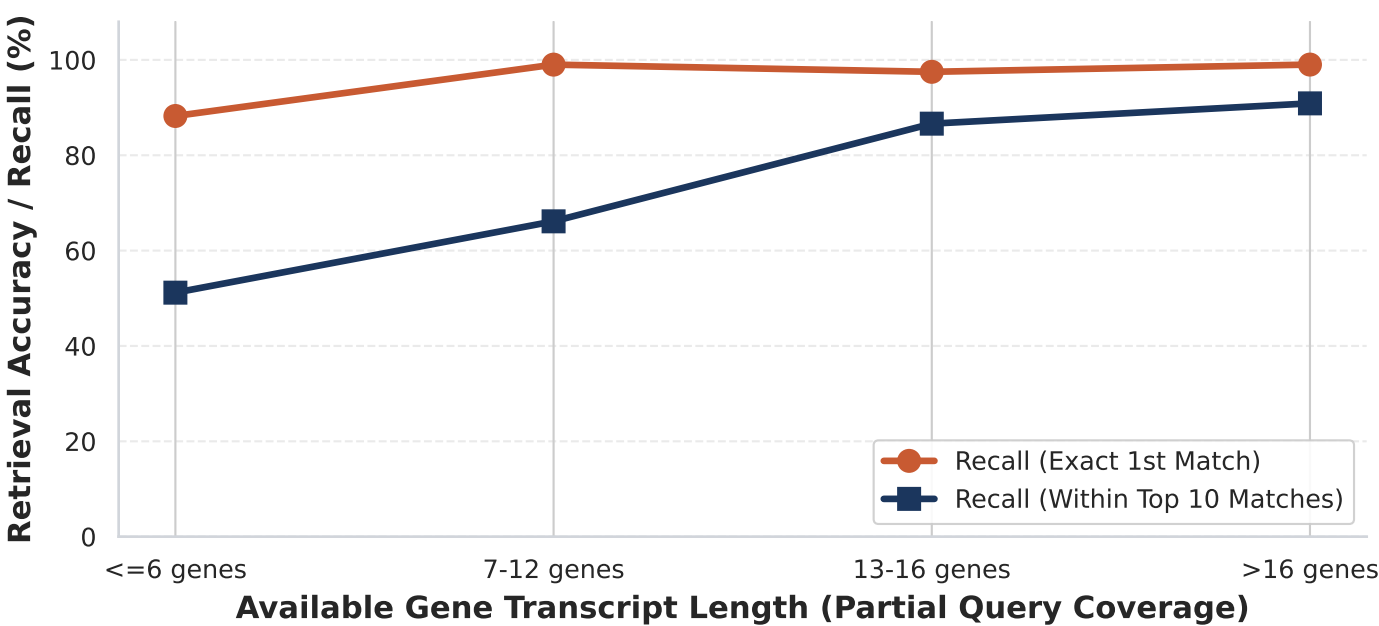
C

Retrieval Rank Accuracy (Exact Matches)



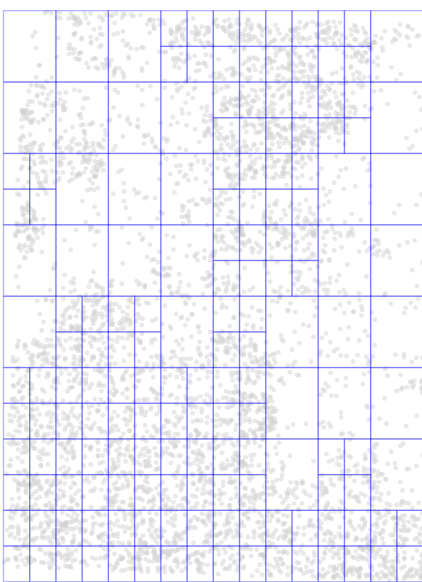
D

Partial Query Search Accuracy

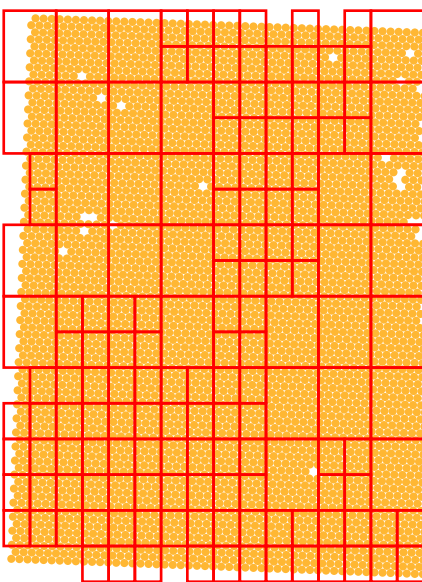


E

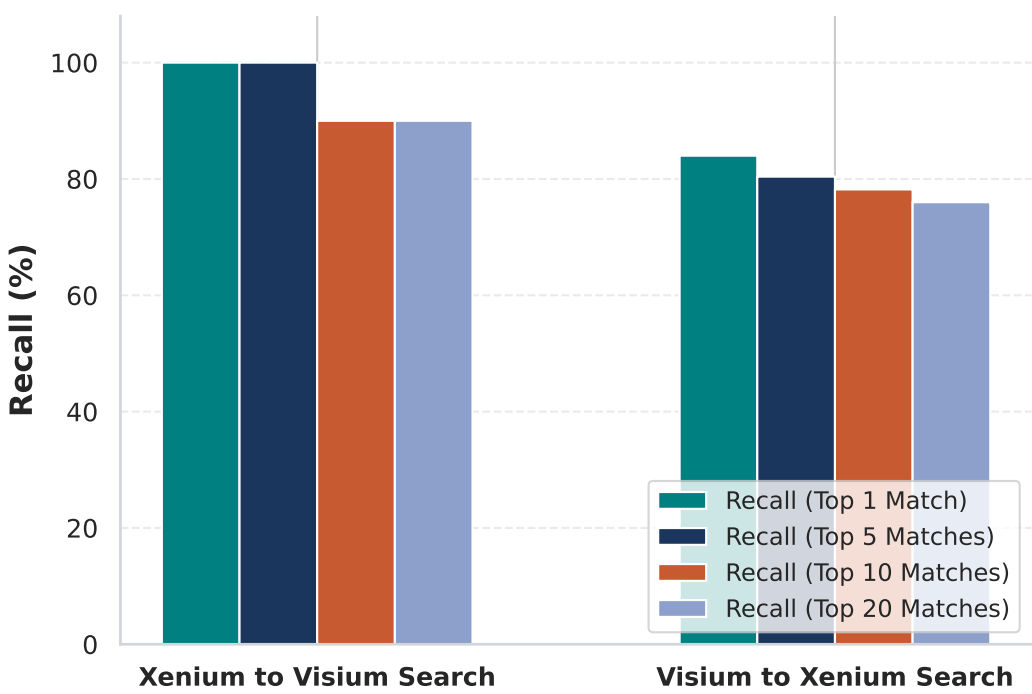
Xenium (166 tiles)



Visium (157 tiles)

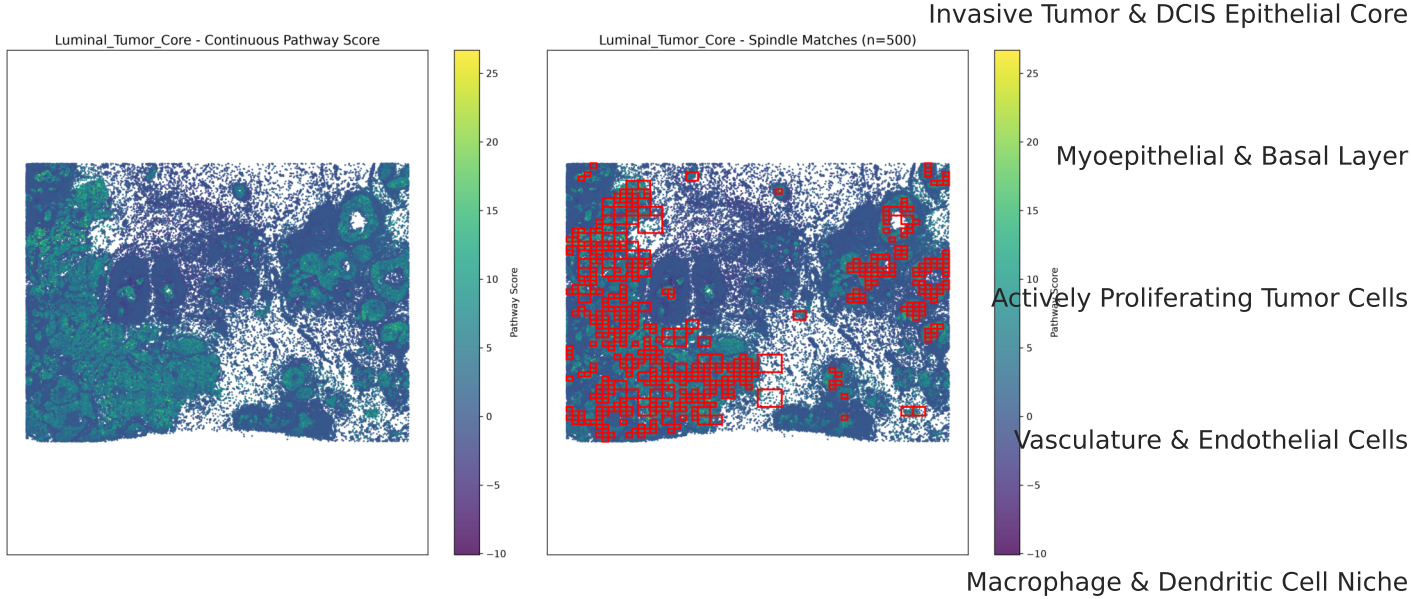


Cross-Modality Retrieval Accuracy (Percentage of true matches found within top candidates)



F

Target Niche Spatial Discovery Map (Luminal Tumor Core)



Pathway Score Enrichment: Background vs. Top 50 vs. Top 10

